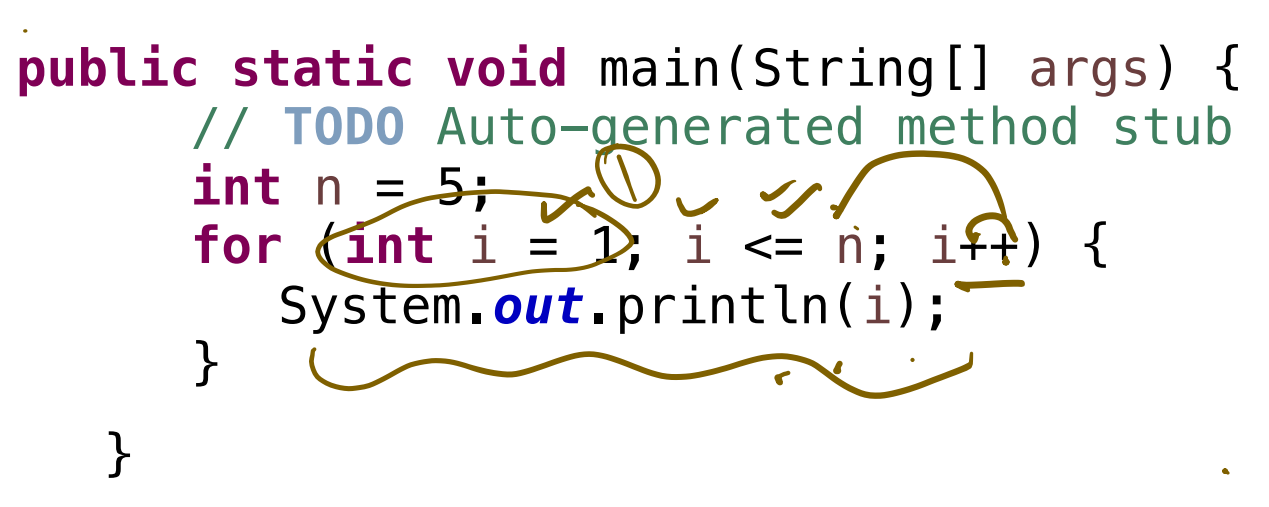
Lec-4
17/10/26 10:12 AM

```
int i=1
while ( i<=n) {
    Self work.
    i++
}
```



```java
public static void main(String[] args) {
      // TODO Auto-generated method stub
      int n = 5;
      for (int i = 1; i <= n; i++) {
         System.out.println(i);
      }

   }
```

i=1	i<=5	
i=2	2<=5	
i=3	3<=5	
i=4	4<=5	1
i=5	5<=5	2
i=6	6<=5	3
		4
		5

Prime-number

N

1 and self divisible

complexity

2 to n-1

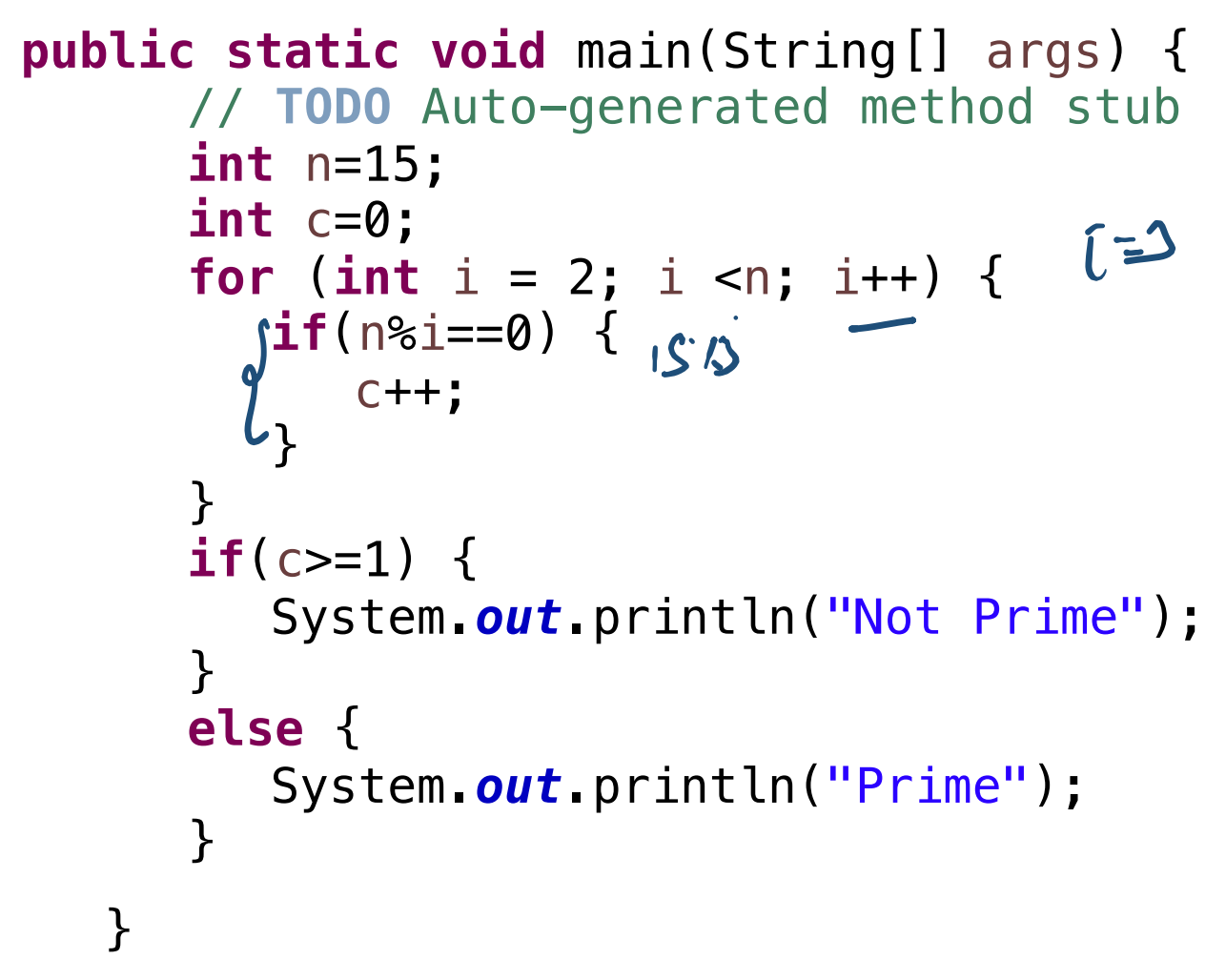
i to

i = 2 to n-1 → n%i==0

c=>2

```
for (int i=2  i<n  i++) {
    if(n%i)==0) {
        c++
    }
}
```

```
int i=2
while(i<n){
    if(n%i==0){
        c++
    }
    i++
}
```



```java
public static void main(String[] args) {
      // TODO Auto-generated method stub
      int n=15;
      int c=0;
      for (int i = 2; i <n; i++) {
         if(n%i==0) {
            c++;
         }
      }
      if(c>=1) {
         System.out.println("Not Prime");
      }
      else {
         System.out.println("Prime");
      }

   }
```

N=15 2 to 14

i=3 15%3 c=1

---

index → 0 1 2 3 4 5 6 7 8 9 10 11

0 1 1 2 3 5 8 13 21 34 55 89 — — -

a b

n times
```
int c=a+b
a=b
b=c
```

N = 6 → 8

① c=a+b 0+1 c=1 a=b b=c

② c=a+b=1+1 =2 a=b b=c

③ c=a+b 2+1=3 a=b→2 b=c→3

④ c=a+b 2+3=5 a=b→3 b=c→5

⑤ c=a+b =3+5→8 a=b →5 b=c→8

⑥ c=a+b 5+8=13 a=b b=c

N 2346

= 2+3+4+6

sum=6+4+3+2

N= 2346 → 234 → 23 → 2 → 0

```
while (n>0) {
    rem=n%10
    sum=sum+rem
    n=n/10
}
```

```
for( ; n>0 ; ){
    int rem=n%10
    sum=sum+rem
    n=n/10
}
```

N = 2346 → Rev → 6432

6 4 3 2

36
63

x×10+2

2348

234×10+8

23×10+8

2348

N = 2436 → 243 → 24 → 2 → 0

0×10+6 = 6 ×10+3
= 63×10+4
= 634×10+2
= 6432

```
rev=0
while (n>0) {
    int rem=n%10
    rev=rev×10+rem
    n=n/10
}
```

60, 36

G.C.D / H.C.F

60 = 1 2 3 4 5 6 10 12 15 20 30 60
36 = 1 2 3 4 6 9 12 18 36

Com (1, 2, 3, 4, 6, 12)

divisor 36) 60 (dividend
36
24) 36 (
24
12) 24 (
24
0

H.C.F ← 12

36) 60 (
36
24

dividend=divisor
divisor=rem

divisor=rem
dividend=divisor

dividend = ~~60~~ ~~36~~ 24   divisor = ~~36~~ ~~24~~ 12

```
while (dividend%divisor != 0) {
    int rem = dividend%divisor
    dividend = divisor
    divisor = rem
}
```

36) 60 (
36
24) 36 (
24
12) 24 (

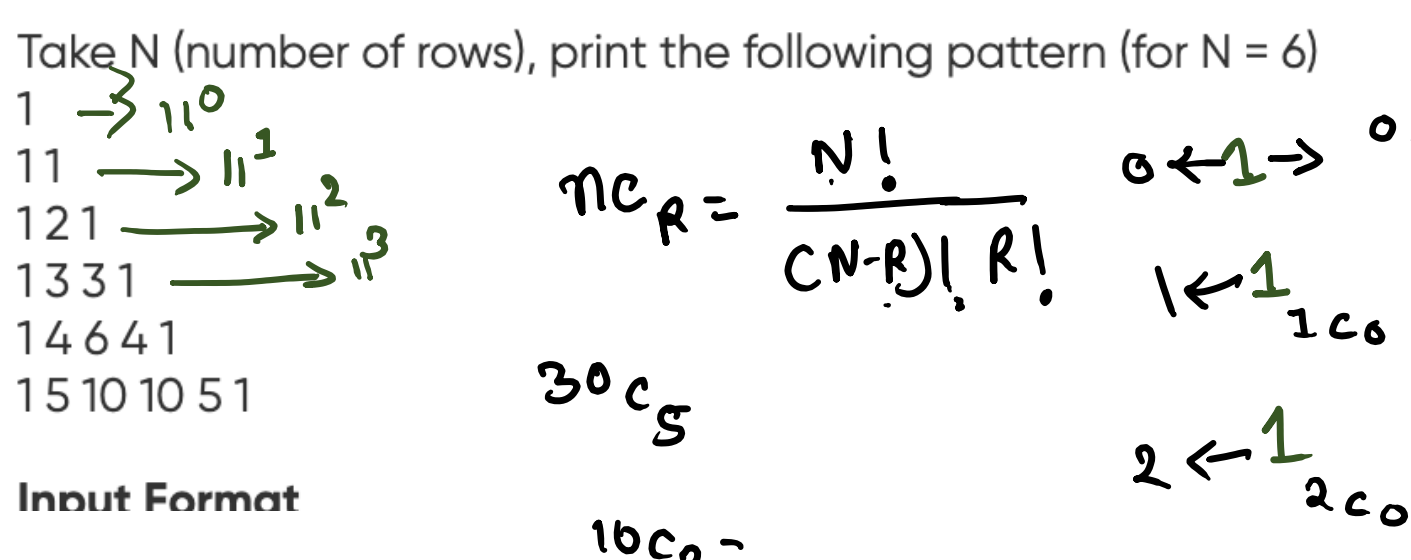
2	378
3	189
3	63
3	21
7	7
	1

2	378
3	189
3	63
3	21
7	7
	1

2, 3, 3, 3, 7

2, 3, 3, 3, 7

```
i= 2 3 4 5 6 7
while(n>1) {
    if (n%i==0) {
        syso(i)
        n=n/i
    }
    else {
        i++
    }
}
```



Take N (number of rows), print the following pattern (for N = 6)
1 → 11^0
1 1 → 11^1
1 2 1 → 11^2
1 3 3 1 → 11^3
1 4 6 4 1
1 5 10 10 5 1

Input Format

$$^nC_R = \frac{N!}{(N-R)!\,R!}$$

$^{30}C_5$

$^{10}C_3$

$5! = 1\times2\times1$

$N! =$

$N! = n\times(n-1)!$

0 ← 1 → $^0C_0$

1 ← 1 $_{1C_0}$ 1 $_{1C_1}$

2 ← 1 $_{2C_0}$ 2 $_{2C_1}$ 1 $_{2C_2}$

3 ← 1 $_{3C_0}$ 3 $_{3C_1}$ 3 $_{3C_2}$ 1 $_{3C_3}$

4 ← 1 $_{4C_0}$ 4 $_{4C_1}$ 6 $_{4C_2}$ 4 $_{4C_3}$ 1 $_{4C_4}$

5 ← 1 $_{5C_0}$ 5 $_{5C_1}$ 10 $_{5C_2}$ 10 $_{5C_3}$ 5 $_{5C_4}$ 1 $_{5C_5}$

n → row
R → i

nCR

$$\frac{n!}{(n-R)!\times R!}$$

$$= \frac{row!}{(row-i)!\,i!}$$

nCR

nCR

$5! = 5\times4!$

$$\frac{P}{Q} \div \frac{R}{S} = \frac{P}{Q}\times\frac{S}{R}$$

$N! = N\times(N-1)!$
$(R+1)! = (R+1)\times R!$
$(N-R)! = (N-R)\times(N-R-1)!$

$$^NC_R = \frac{N!}{(N-R)!\times R!} \quad — ①$$

$$^NC_{R+1} = \frac{N!}{(N-(R+1))!\times(R+1)!} \quad ⑪$$

⑪ ÷ ①

$$\frac{^NC_{R+1}}{^NC_R} = \frac{\cancel{N!}}{(N-R-1)!\times(R+1)!}\times\frac{(N-R)!\times R!}{\cancel{N!}}$$

$$\frac{^NC_{R+1}}{^NC_R} = \frac{(N-R)!\times \cancel{R!}}{(N-R-1)!\times(R+1)\cancel{R!}}$$

$$\frac{^NC_{R+1}}{^NC_R} = \frac{(N-R)\cancel{(N-R-1)!}}{\cancel{(N-R-1)!}\times(R+1)}$$

$$\frac{^NC_{R+1}}{^NC_R} = \frac{N-R}{R+1}$$

$$^NC_{R+1} = \left[\frac{(N-R)}{R+1}\right]\times {^NC_R}$$

$$\left[\frac{row-i}{i+1}\right] nCR$$